Understanding Habits: The Enactive Perspective

Siddharth Chundru

Carleton College

Abstract

Habits have long been a focal point in psychological research, and have come to be characterized as automatic behaviors formed through repetitive stimulus-response associations and activated by environmental cues. This view, heavily influenced by dual processing theory and associationist principles, operates from within a mindless-mindful dichotomy that oversimplifies the complexity of actual habitual actions. Recent empirical evidence, which indicates that habits exhibit flexibility, adaptability, and can involve conscious deliberation, has come to challenge this perspective. This paper critiques the traditional cognitivist understanding of habits and explores the enactivist approach as a more comprehensive alternative framework. Enactivism, rooted in the organicist tradition, views habits as dynamic, self-organizing patterns emerging from an organism's embodied interactions with its environment. By positioning habits at the core of cognitive processes, enactivism emphasizes their fundamental role in shaping intelligent behavior. The paper finally reviews computational models employed by enactivists, demonstrating how these models provide valuable insights into the formation and maintenance of habits.

Keywords: habits, dual processing theory, enactivism, cognition, sensorimotor engagement, computational modeling

Section 1: Introduction

Habits play an important role in shaping human behavior, influencing actions ranging from split second decisions to complex decisions. Traditionally, psychology has characterized habits as automatic behaviors formed through repetitive stimulus-response associations, activated by environmental cues (Volpp & Loewenstein, 2020). This view has emerged from the dual processing theory, which categorizes cognitive processes into automatic and controlled systems (Saling & Phillips, 2007). According to this perspective, habits are mindless automatisms performed without conscious deliberation or intentional goal pursuit (Gardner, 2015). However, recent empirical evidence challenges this simplistic characterization. Studies have shown that habits often exhibit flexibility, adaptability, and can even involve conscious control and intentionality (Wood & Rünger, 2016) (Moors & De Houwer, 2006). Which suggests that the traditional cognitivst views over simplify the complex nature of habitual behavior.

This paper begins by exploring the evolution of the concept of habit, contrasting the associationist perspective—which views habits as rigid and automatic—with the organicist tradition that sees habits as flexible, dynamic structures intertwined with an organism's sensorimotor capabilities and environment (Barandiaran & Di Paolo, 2014). Then, I introduce enactivism, an approach rooted in the organicist tradition, which states that cognition arises through dynamic interactions between an organism and its environment, and emphasizes the importance of embodied action as well as the fundamental intertwining of perception and action (Varela, Thompson, & Rosch, 1991).

By providing an extensive overview of enactive theories and reviewing computational models employed by enactivists, I aim to provide a more accurate understanding regarding the actual nature of habits. I also aim to advocate for a paradigm shift in psychological research, by presenting a perspective that moves beyond the limitations of the mindless-mindful dichotomy, and highlights the true significance of habits in cognition.

Section 2: Dual Processing Theory

Habits have remained a historically important topic of study within psychology. From early behaviorist conceptions to more contemporary computationalist and enactive perspectives there have existed a variety of perspectives on habits. Today however, they are generally characterized as "behaviors automatically activated by recurring environmental cues", and are thought to be formed through repetitive behaviors and "learned stimulus-response associations" (Volpp, K. G., & Loewenstein, G., 2020). It is mostly agreed that once established, habitual actions are performed automatically, without conscious deliberation and are not driven by the intentional pursuit of specific goals (Gardner, 2015).

Habits have come to be defined by modern psychology in this manner mainly due to the influence of the dual processing theory (Saling, L. L., & Phillips, J. G., 2007), first introduced by Wason, P. C., & Evans, J. ST. B. T. (1974) . According to this theory, cognitive processes can be characterized as either 'automatic' or 'controlled'. Automatic processes are defined as being fast, free of errors, effortless, stereotypic and autonomous. Generally, they operate without conscious

awareness, often functioning involuntarily or unconsciously, and are not limited by attentional capacity, occurring alongside other cognitive processes without interference. While controlled processes on the other hand are more flexible and better suited for dealing with new situations, but also more effortful, slow, and prone to errors (Fabio, R. A., Caprì, T., & Romano, M., 2019).

Automatic processes are typically understood as products of pre-existing mental frameworks where concepts are associated with valences (positive or negative emotional charge) and are seen as the mechanisms behind habitual responses, which generate spontaneous unintentional action, whereas controlled processing is used when making intentional or thoughtful actions, and involves conscious and effortful retrieval of information from memory (Fabio, R. A., Caprì, T., & Romano, M., 2019).

Section 3: The Evolution of The Concept of Habit

In order to understand how the dual processing theory emerged and rose to prominence in modern psychology, we look at Brendarian and Di Paolo (2014) who traced the evolution of the concept of habits in the western intellectual history. In their analysis, they identified a major bifurcation with two prominent perspectives: the organicist and the associationist views. Associationists have tended to view habits as rigid, mechanistic, atomistic, and automatic, while organicists view habits as flexible and dynamic structures that are fundamentally tied to how an organism's sensorimotor capabilities relate to its environment. The dual processing theory gained popularity when cognitivism, which sought to describe mental processes as akin to computer programs being instantiated on a symbol processing machine, was the dominant theoretical

framework in psychology (Garnham, A., 2019). Brendarian and Di Paolo (2014) have traced this cognitivist view of habits back to the associationist school of thought.

According to Brendarian and Di Paolo (2014), associationists tend to view mental phenomena as resulting from the association or combination of simple atomic elements. According to the associationist principle, event A will occur given event B if event A has repeatedly occurred following event B in the past. These events are generally viewed to be mental states (often conceptualized as patterns of neural activation) stemming from sensory experiences (Barandiaran & Di Paolo, 2014).

Brendarian and Di Paolo (2014) credit David Hume with introducing habits or 'associations' as the fundamental mechanisms underlying mental processes. In *A Treatise of Human Nature* (Hume, 1739), Hume argues that all ideas, which serve as the primary principle of mental organization (i.e., atomic primitives), are ultimately derived from sensations. He then introduces the law of habit as the principle factor behind this mental organization. He claims that habits enable the mind to associate ideas by connecting experiences in terms of cause and effect. And Brendarian and Di Paolo (2014) credit David Hartley with bridging Hume's ideas with psychology. In his book *Observations on Man* (Hartley, 1749) Hartley developed the concept of 'association of ideas' to propose that both mental and physiological processes follow the principle of association. This allowed Hartley to expand on Hume's original idea to a broader range of psychological phenomena, including memory, language, psychological development, and emotions.

Brendarian and Di Paolo (2014) claim that Hume's and Hartley's original ideas continued to influence western theories of mind for the next two centuries and acted as the precursor for Behaviorism. Behaviorists rejected introspection, which had up til the advent of behaviorism been a popular tool used for psychological analysis, as a legitimate scientific tool (Zilio, D., 2016). Instead, they limited the scope of their analysis to observable behavior, stating that their key aim was to control and predict behavior (Watson, J. B., 1913) (Barandiaran, X. E., & Di Paolo, E. A., 2014).

This focus on observable behavior resulted in behaviorists contributing to the associationist tradition by redefining habits in two major ways: First, by refusing to acknowledge the role of any intermediary psychological or neurological processes, they caused the externalization of all internal process of association into explicit stimulus-response patterns, and second by creating a rigid mathematical relationship between external features and observable behaviors (Barandiaran, X. E., & Di Paolo, E. A., 2014).

The behaviorists' refusal to consider or even acknowledge internal processes resulted in the cognitivist revolution of the 1950s (Miller, G. A., 2003). The focus of study shifted from behaviors and associations to mental representations and computation (Barandiaran, X. E., & Di Paolo, E. A., 2014). While cognitivists did acknowledge that habits or 'automated behaviors' develop through extensive practice and repetition, they were no longer foregrounded as foundational to the practice of psychology (Shiffrin, R. M., & Schneider, W., 1977) (Barandiaran, X. E., & Di Paolo, E. A., 2014). Instead, habits were seen as optimizing mechanisms: once behaviors become habitual, they can be executed with little to no conscious

effort, freeing up cognitive capacity for more complex or novel tasks (Shiffrin, R. M., & Schneider, W., 1977).

Section 4: Mindless-Mindful Dichotomy

Today, due to the influence of associationism and the dominance of the dual processing theory, habits are typically portrayed as strict mindless automatisms, and seen as automatic, rigid responses devoid of conscious thought (Evans & Stanovich, 2013) (Kahneman, 2011). The cognitivist view has confined the discourse on habits within a mindless-mindful dichotomy, i.e. behaviors are either entirely automatic or fully deliberate (Ramírez-Vizcaya, S., & Froese, T., 2019). However, given the complex nature of habits, this binary categorization has increasingly been criticized and viewed as overly simplistic (Ramírez-Vizcaya, S., & Froese, T., 2019). Empirical studies have shown that habits tend to display a surprising degree of fluidity, adaptability, and context-sensitivity (Wood & Rünger, 2016) (Moors & De Houwer, 2006). And, there is a mounting degree of evidence that challenges the notion that habitual behaviors lack conscious control or intentionality, creating a need to reconsider the simplistic definition of habits offered by cognitivists (Ramírez-Vizcaya, S., & Froese, T., 2019).

Ramírez and Froese (2019) have observed this conflict surrounding the nature of habits across various disciplines including neuroscience (connectivism), organizational sciences, and philosophy of mind. In organizational sciences, routines are typically regarded as rigid schemas or as predictable and ingrained patterns of behavior (habits) that arise through repetition (Nelson & Winter, 1982) (Cohen & Bacdayan, 1994). However, studies show that routines can change

and evolve intrinsically (without being driven by external factors) as a result of intentional thought and reflection (Feldman & Pentland, 2003) (Pentland & Feldman, 2005). Leading scholars, like Feldman and Pentland (2003), to argue that organizational routines are not static but change dynamically, being capable of adapting to new contexts through mindful engagement.

The mindful-mindless dichotomy has also been a point of contention in philosophy (Ramírez-Vizcaya, S., & Froese, T., 2019). Gilbert Ryle (1949) draws a distinction between habits and intelligent thought, suggesting that habitual actions operate without conscious deliberation and are less flexible than actions requiring intelligence. According to him, habits are automatic processes that function independently of rational thought. John McDowell (1994), on the other hand, challenges this view stating that rational and mindful human activity necessarily involves habits. According to him, habits are not just repetitive behaviors, but are informed by and in turn inform our rational capacities, and therefore fundamental to conscious understanding and intelligent action.

Hubert Dreyfus (2002) also presents a perspective that challenges Ryle's (1949) ideas about habits. Dreyfus (2002) analyzes the non-conscious, embodied nature of certain skilled behaviors, and argues that many complex actions which are often perceived as outcomes of deliberate rational thought, are actually deeply rooted in embodied 'know-how' and operate below the level of conscious awareness. He argues that embodied habits are central to intelligent behavior, claiming that expertise involves a level of intuitive, non-reflective action which occurs when an individual is fully absorbed in a task.

To move beyond this mindless-mindful dichotomy, it is essential to recognize the dynamic and context-sensitive nature of habits. Recognizing the flexibility of habits as well as the fundamental role perceptual skills play in habitual behaviors, allows us to view habits not as merely automatic response but as also both influenced by and influencing environmental interactions and intentionality (Clark, A., 1997) (Merleau-Ponty, M., 1962). Organicist views, that emphasize the complexity and self-organizing nature of habits, are better suited to address this interplay between automaticity and mindfulness in habitual behaviors.

Section 5: Enactivism

Enactivism, a product of the organicist tradition, is a biologically inspired approach that recognizes the dynamic and flexible nature of habits by emphasizing the embodied, situated, and emergent aspects of cognition. Shifting focus to the enactive perspective can allow us to move beyond the mindless-mindful dichotomy, and adopt a more comprehensive view of habits.

Barandiaran and Di Paolo (2014) identify organicism as the other major trend that emerged in parallel with the associationist tradition. The organicists tend to view habits as 'ecological' and 'self organizing'. In contrast to the associationist viewpoint, which typically considers habits to be more passive– results of repeated associations between pre-existing entities such as mental states, rewards and stimuli, the organicist tradition considers habits to be active processes (Barandiaran, X. E., & Di Paolo, E. A., 2014) . In this framework, habits are both the cause and the effect of their own enactment—they shape themselves through their performance (Barandiaran, X. E., & Di Paolo, E. A., 2014). This means that habits create their own identity or

"principle of individuation" because they are not merely imposed by external factors but develop from the organism's own activities (Barandiaran, X. E., & Di Paolo, E. A., 2014). Furthermore, the organicist perspective considers habits to be part of a flexible and adaptive equilibrium ("plastic equilibrium") that involves the entire organism. This includes other habits, the physical body, and the environment ("habitat") they collectively influence and co-create (Barandiaran, X. E., & Di Paolo, E. A., 2014). In essence, habits are seen as dynamic patterns that emerge from and contribute to the organism's ongoing interaction with its environment.

Influenced in part by organicist thinkers like Maurice Merleau-Ponty, John Dewey and William James, enactivism emerged as a viewpoint that stands in contrast to cognitivism (Barandiaran, X. E., & Di Paolo, E. A., 2014). According to the enactive perspective, cognition arises through dynamic interactions between an organism and its environment, rather than strictly through machine-like information processing based on internal representations. (Varela, F. J., Thompson, E., & Rosch, E., 1991). The enactive view holds that knowledge is enacted through embodied action, with perception and action being fundamentally intertwined (Varela, F. J., Thompson, E., & Rosch, E., 1991).

From an enactive standpoint, habits are not pre-programmed, automatic responses but are emergent patterns of behavior resulting from an organism's history of interactions with its environment (Pérez-Verdugo, M., & Barandiaran, X. E., 2023). They are self-sustaining and adaptive, allowing organisms to navigate their environment effectively, and are considered fundamental to cognition, shaping and being shaped by ongoing sensorimotor engagements (Ramírez-Vizcaya, S., & Froese, T., 2019).

According to enactivists, habits are at the core of mental activity. They disagree with the cognitivist characterization of habits as automatic processes peripheral to intelligent thought. In enactivism, habits are seen as fundamental features that provide the stability and continuity necessary for coherent action in a complex world (Barandiaran, Di Paolo, & Rohde, 2009). Enactivists believe that habits constitute a form of embodied know-how, which enables organisms to anticipate and adapt to environmental changes.

Additionally, enactivists believe that habits are fundamentally tied to an organism's sensorimotor capabilities (Varela, Thompson, & Rosch, 1991). Habits are classified as regularities in perception and action that develop through repeated interactions (Di Paolo, Buhrmann, & Barandiaran, 2017). They are not believed to be stored internally, but instead 'enacted' through continuous coupling with the environment (O'Regan & Noë, 2001). For example, learning to ride a bicycle involves developing sensorimotor habits that integrate balance, pedaling, and steering through embodied practice (Noë, 2004).

Radical enactivism, a branch of enactivism characterized by its refusal to acknowledge that any form of inner mental representation plays a part in cognition, places particular emphasis on the relationship between bodies and habits (Hutto & Myin, 2013). In this view, the body and its interactions embody the history of habits, with cognition being an emergent property of these embodied engagements (Gallagher, 2005). The body is not just a passive vessel but an active participant that carries traces of past interactions and influences present behavior (Hutto & Myin,

2017). Habits are both fundamentally shaped by and shape the organism's embodied history (Chemero, 2009).

Enactivists have even extended their theories of habits into the realm of social interaction. Enactive theories argue that socially dependent habits emerge from participatory sense-making, where social cognition is co-constructed through interpersonal engagements (De Jaegher & Di Paolo, 2007). Social habits are seen as patterns of interaction that become regularized through shared practices, such as conversational norms or cultural rituals (Froese & Di Paolo, 2011).

Empirical models can serve as valuable exploratory tools for the enactive study of habits. Historically, enactivists have successfully employed a range of tools including neural networks, genetic algorithms, and dynamic systems theory in order to construct their models (Beer, R. D., 1995) (Di Paolo, E. A., 2003). The next section will be a literature review of some empirical models, which use many of the aforementioned tools, used by enactivists to study habits. These models are valuable as they can provide some key theoretical insights into various aspects of habitual behavior.

Section 6: Literature review:

Sensitivity to social contingency or stability of interaction? Modelling the dynamics of perceptual crossing (Di Paolo, E. A., Rohde, M., & Iizuka, H., 2008)

Based on the idea that behavioral patterns are not rigid mechanisms within an agent's

cognitive architecture but are shaped through interaction dynamics, Di Paolo, Rohde, and Iizuka (2008) developed a minimal model of behavior. They used their model to explore the dynamics of social interaction and *perceptual crossing*, the phenomenon where multiple individuals actively engage in a reciprocal exchange of sensory feedback (Deschamps, L., Lenay, C., Rovira, K., Le Bihan, G., & Aubert, D., 2016). Despite certain limitations regarding its potential for accurately explaining human behavior, their model successfully showed that sensorimotor habits arise from ongoing interaction between agents and demonstrated how coordinated behavior can emerge through mutual perception.

In the model they developed, Di Paolo, Rohde, and Iizuka (2008) showed that simulated agents can develop coordinated behaviors through mutual interaction in a minimal virtual environment. They optimized continuous time recurrent neural networks (CTRNN), a type of neural network, using genetic algorithms for the following task: Agents were placed in a one dimensional environment amidst static objects, moving shadows of other agents, and a mutual partner agent. The partners were tasked with identifying and spending as much time as possible as close to each other as possible while not being trapped by the other features of the environment.

As the agents engaged with their environment, the global interaction dynamics, i.e., the overall patterns of interaction that emerge from the continuous and interconnected feedback between the agents' movements, sensory input, and the behavior of the environment, shaped their ability to distinguish between different entities. Rather than relying on internal cognitive mechanisms to make distinctions, the interaction process itself led to behavioral stability. That is to say, rather than relying on pre-programmed responses, the agents learned to differentiate

entities through the cumulative effects of their interactions. When agents engaged with responsive entities (such as other agents), they experienced a dynamic, two-way interaction, whereas non-responsive entities (like shadows) did not provide feedback. Over time, the patterns of feedback and coordination from the environment allowed the agents to develop behaviors that were fine-tuned to different types of entities. This supports the notion that behavioral patterns can emerge from collective dynamics, rather than being the result of individual cognitive architectures.

Contingent (reciprocal) interactions between partners allowed agents to successfully differentiate between partners and unresponsive entities (shadows), despite receiving similar sensory input from both. Di Paolo, Rohde, and Iizuka (2008) argue that the stability of the partners' mutual interaction was key to solving the task, as mutual perceptual activity was necessary for coordinated behavior to take place. Only when both agents were mutually responsive to each other's actions did their interactions stabilize and result in coordinated behavior. When one agent perceived and responded to the movements of another, it reinforced its perception that it was interacting with a live, responsive partner rather than a static object.

Di Paolo, Rohde, and Iizuka (2008) also found that the stability of interactions played an important role in an agent's ability to make distinctions. They found that agents did not distinguish their partners from non-responsive shadows using direct perceptual clues, but rather through the stability of interactions. While interactions persisted when both partners engaged in mutual scanning, interactions with shadow images would eventually break down due to the lack of contingency– which allowed agents to make distinctions.

The findings of this paper highlight the multi-faceted nature of sensorimotor behavior and underscore the importance of an approach that looks beyond individual capabilities and the bounds of the agent's cognitive architecture to instead also consider embodied and emergent-interactive phenomena. It is not a pre-coded algorithm but the interaction process itself that is key to understanding how behaviors emerge and persist over time. Tasks that seem intuitively difficult, such as distinguishing between two entities that move identically, such as the partner and its shadow image, become trivialized when the effects emerging from the mutual search for each other are taken into consideration.

While cautioning that these findings do not necessarily represent exactly the underlying mechanisms behind human behavior, the authors do claim that these findings represent one possible solution that humans could be using for this specific task. The authors go on to claim that this model's success in modeling interactive dynamics effectively showcases the strengths of using an algorithmic approach with minimally complex agents for modeling in enactive research. While not being able to capture the full scope of human beings, they can help better understand particular and limited aspects of human sensorimotor behavior.

Di Paolo, Rohde, and Iizuka (2008) were able to successfully model the emergence of dynamic yet stable patterns of sensorimotor behavior using an algorithmic approach, which paved the way for modeling enactive habits using similar techniques. Habits under enactivism, also being dynamic yet stable patterns of sensorimotor behavior, have also been modeled with great effectiveness using similar techniques. Some of these modeling efforts will be presented in the subsequent subsections.

Toward Spinozist Robotics: Exploring the Minimal Dynamics of Behavioural Preference (Iizuka, H., & Di Paolo, E. A., 2007)

In a prior paper, Iizuka and Di Paolo (2007) developed a minimal model of behavioral preference to illustrate the notion that a behavioral disposition towards a specific action (a habit) is not contained within an agent's cognitive architecture, but is instead a constraining factor of behavior that is itself shaped by said behavior. Their model was able to capture this relationship between sensorimotor habits and behavioral tendencies, and they successfully produced an agent that displayed the processes of habituation and re-habituation. Despite this success, the authors acknowledged certain limitations, particularly regarding the degree of autonomy in the agent's actions, which will be discussed later in this subsection.

In this model, Iizuka and Di Paolo (2007) demonstrated that a simulated agent develops behavioral preferences through a combination of homeostatic adaptation and environmental interactions. The agent, similarly to the model in Iizuka and Di Paolo (2008), was evolved using a genetic algorithm and was tasked with responding to visual stimuli by moving towards target lights while maintaining its internal variables within two designated homeostatic bounds. These bounds were linked to different behaviors, with the agent's neural dynamics being contained within high-dimensional "boxes" that corresponded to these behaviors.

When the agent's neural dynamics remained within a particular bound, it displayed behaviors associated with that particular bound. As the agent interacted with its environment over time, it developed preferences (or habits) based on the history of those interactions. When presented with two equally viable behavioral options, the agent's behavior was neither random nor

transient, but rather stable and durable, indicating habituation. Moreover, when faced with changes in its internal or environmental conditions, the agent demonstrated the capacity to switch between behavioral preferences, adapting its response, and thereby demonstrating rehabituation.

The authors characterize these habits as dynamic because, while the agent's preferences were persistent, they were not permanent. Given that the agent lacked internal representations, the changing dynamics of its interactions best explains why it was able to exhibit stable but flexible behavioral preferences. This finding can be interpreted as evidence that rather than through pre-programmed mental states and internal representations, habitual behaviors can arise through dynamic interactions between internal and external variables.

One interesting finding was that, in some cases the agent continued searching for a target light that was no longer present while avoiding an alternative, available target light. Iizuka and Di Paolo (2007) draw a parallel between this phenomenon and the persistence observed in object permanence tasks in 7 to 10 month old infants. Given that the simple dynamic mechanisms used in their model were sufficient to replicate the error patterns seen in infants (which gradually disappear with age), this model lends credence to the idea that motor memories form the basis for perseverance and offers a plausible sensorimotor explanation for the development of object permanence (Smith, L. B., & Thelen, E., 2003).

While the authors argue that the model's ability to capture these aspects of behavioral preference and habit formation highlights the strengths of employing a dynamic systems approach (DST) in enactive research, they did recognize one particularly prominent limitation

of their model with regards to its capacity for autonomy. The model's reliance on two fixed internal homeostatic regions served to limit the agent's autonomy, leading the authors to suggest that a more nuanced alternative approach might involve moving gradients of plasticity instead of rigid pre-determined bounds. Such an approach would allow for the spontaneous formation of stable regions that reflect the history of the agent's interactions, not only enabling the agent to switch between externally provided goals but also allowing it to create its own goals. This is particularly problematic because one could argue that habitual behaviors in autonomous real-life agents require mental states. Therefore, in the next subsections, I will discuss models that involve agents with greater capacity for autonomy.

Homeostatic plasticity in robots: From development to operant conditioning to habit formation. (Barandiaran and Di Paolo, 2010)

Given the limitations of the previous model in demonstrating autonomous behavior, Barandiaran and Di Paolo (2010) developed a new model that focused on enhancing plasticity and adaptability. In this model, they used a homeostatically plastic CTRNN whose connectivity was modified using a presynaptic Hebbian rule when the firing rate of neurons went above or below specified bounds. This setup allowed the neural architecture to remain flexible while maintaining stability. In this model, behavioral sensorimotor patterns became linked to emergent regions of stable synaptic plasticity. Since plasticity was activity-dependent, variations in sensorimotor history enabled the generation, reshaping, and switching between regions of stability, allowing for adaptable behaviors based on the agent's interactions with its environment.

In this model, the agents were artificially evolved using an operant conditioning task, which involved selecting between differently colored food sources with varying levels of benefit or harm. The key aspect of this task was that the changing profitability of food sources over time required the agent to adapt its behavior. Barandiaran and Di Paolo (2010) demonstrated that the agents were able to learn the association between colors and profitability, as well as adapt accordingly as things changed. This dynamic adaptability allowed the agent to maintain, switch between, and, most significantly, spontaneously generate new habits (that were not present initially when evolved).

An important finding of their study is that the agents exhibited emergent behaviors that were not pre-programmed or evolved. For instance, when presented with two equally profitable food sources, the agent developed a preference for choosing food on its right. This behavior was not an innate preference but formed through the agent's interaction history during the experiment and became evident after several trials. When the authors tested different conditions where profitability wasn't a criterion (i.e., both food items were equally beneficial or harmful), the agent's history of interactions still resulted in the formation of specific preferences or habits, showing the impact of sensorimotor correlations on behavior.

While the agents in this model showcased a wide range of autonomous behaviors, the authors acknowledge that the high dimensionality of the homeostatically plastic system made it very complex and hard to analyze in detail, and they suggested that further testing and exploration was required inorder to better understand the intricacies of the model. In the next section, I will present an alternative model with agents that display spontaneous habitual behavior while being easier to analyze.

Modeling habits as self-sustaining patterns of sensorimotor behavior (Egbert, M. D., & Barandiaran, X. E., 2014)

Given the increasing complexity of habit modeling efforts by enactivists, Egbert and Barandiaran (2014) created an alternative framework that allows for easier analysis while still generating autonomous and dynamic habitual behavior. They referred to this model as operating on a meso-scopic level of abstraction, meaning it functions at a level above microscopic details about individual neurons and below macroscopic "discretized and individualized stimulus and response units." This model focuses on the dynamics of sensorimotor coordination where behavioral patterns emerge from continuous sensorimotor loops, without relying on explicit neuronal assumptions or isomorphic mappings between representations and the environment.

To achieve this meso-scopic level of abstraction, Egbert and Barandiaran developed the Iterant Deformable Sensorimotor Medium (IDSM), a tool that couples with an agent's sensorimotor body and modulates its motor output based on its sensorimotor state, rate of sensorimotor change, and the current state mapping. The IDSM continuously creates and modifies nodes, which are essentially records of the agent’s sensorimotor dynamics. Each node consists of a sensorimotor state, the rate of change in that state, and a weight factor. As the agent performs actions and its sensorimotor state changes, new nodes are added, and the system uses a self-reinforcing mechanism: when the agent encounters a similar sensorimotor state, the IDSM repeats past behaviors, with repeated patterns of behavior becoming reinforced, making recent and frequent behaviors more likely to be repeated in the future.

The authors were able to observe a variety of habitual behaviors by initializing the IDSM-based agent using different training conditions. The range of observed behaviors includes recreation of previous behaviors, selective training of functional habits, and the emergence of new behaviors that were not present during the initial training phase (autonomous habit generation). The system's strength lies in the way repeated sensorimotor actions stabilize into specific behavioral patterns. For instance, during the initial training phase, an agent might be driven to perform specific behaviors, such as approaching a light, and this behavior is recreated in later free action phases.

One of the most significant aspects of the IDSM is its ability to generate self-sustaining behavioral patterns—or habits—without the need to be pre-programmed. The IDSM allows the agent to autonomously form and maintain habits through repeated sensorimotor trajectories. The IDSM is an important advancement over earlier models, as it shows how habits dynamically respond to and emerge from an agent's interaction history. By inducing reinforcement through repetition, the IDSM allows the agent to stabilize behaviors in a continuous, dynamical framework.

Egbert and Barandiaran highlight the fragility of habitual behaviors, which rely on sustained interaction with the environment to continue existing, by likening these self-maintaining behavioral habits to metabolic processes found in living organisms. Just as metabolic processes are dissipative structures that rely on continuous external energy sources to maintain themselves, habits are precarious structures that depend on repeated sensorimotor contingencies within the environment to persist. Without repetition-induced reinforcement, habits gradually diminish and even eventually disappear. They use this analogy to

The findings of this study also show that habits can vary in their adaptability, with some habits being more adept at persisting than others (depending on the conditions of the agent's environment). This is because habits in this model can interact either competitively or cooperatively with each other, leading to some habits persisting or existing over others. For example, the authors observed that during some trials, certain habits prevented the agent from exploring particular regions of its environment, thereby disallowing the formation of certain habits that were present in other trials.

Egbert and Barandiaran move away from traditional notions of internal representations by reframing habits within the context of the relationship between an agent and its environment. The IDSM contains a history of the agent's sensorimotor trajectory and flexibly adapts to new interactions, allowing for the emergence of complex, adaptive behaviors without the need for a one-to-one mapping between internal representations and external stimuli, allowing the claim that habitual behavior arises out of a dynamic interaction between the plastic IDSM, the agent's body, and the environment.

This model can also provide valuable theoretical insights to enactivists regarding the role repetition plays in habit formation. Repetition is the core mechanism behind the formation and persistence of habits in the IDSM. The IDSM reinforces sensorimotor states by revisiting nodes associated with past behaviors. This leads to the formation of stable sensorimotor patterns, or habits, that the agent is likely to repeat. Over time, these habits become self-maintaining, meaning they persist autonomously as long as the environmental conditions allow for their reinforcement.

Egbert and Barandiaran's IDSM model demonstrates how sensorimotor dynamics—operating at a meso-scale—can autonomously generate and maintain behaviors in a way that mimics the self-organizing properties of living systems. This framework can be seen to favor radical enactivism as it allows for modeling the emergence and persistence of habits by focusing solely on self-reinforcing sensorimotor patterns, without relying on representationalist frameworks. In contrast, earlier models by Di Paolo arguably allowed for the incorporation of mental states as causal mechanisms of habit generation alongside dynamic organism-environment interactions.

Different Forms of Random Motor Activity Scaffold the Formation of Different Habits in a Simulated Robot (Zarco, M., & Egbert, M., 2019)

Given the success of the IDSM framework developed by Egbert and Barandiaran (2014), Zarco and Egbert (2023) conducted further experiments using the same model which showed that even slight variations in the model led to significant plastic changes, that resulted in different sensorimotor behaviors. By slightly tweaking the parameter settings in the random motor activity phase, they induced a range of different behavioral outcomes, including static, monotonic, and oscillatory patterns.

Zarco and Egbert (2023) built upon the same upon the same theoretical underpinnings of Egbert and Barandiaran (2014), with their central premise being that habits are formed through the repeated enactment of sensorimotor regularities and maintained by plastic changes in the behavior-generating medium. According to them, habits, or sensorimotor autonomies, emerge as self-maintaining organizations within a biological individual. They used this study

to provide support for the idea that biological autonomy and sensorimotor autonomy are linked through evolutionary pressures that favor the formation of functional habits.

Zarco and Egbert (2023) used a simple robot controlled by an IDSM, which modulated the robot's motor output based on its sensorimotor state, the rate of sensorimotor state change, and the current state mapping. As in the previous study, the IDSM continuously created and modified nodes, which represent records of sensorimotor dynamics. Each node includes the sensorimotor state, its rate of change, and a weight factor. The robot's behavior evolved as these nodes were created and reinforced through repetition, leading to the emergence of self-sustaining sensorimotor patterns.

The experiment focused on varying the parameters of random motor activity in the agent, specifically the probability of setting a new random vector (R) and the variance of the normal distribution from which R was drawn. The robot operated in a one-dimensional periodic environment (a closed one dimensional loop), and its behavior was classified into three types—static, monotonic, and oscillatory—based on its sensorimotor trajectories. Low values of probability (p) led to monotonic behaviors, where motor values were maximized. While, sporadic changes in R, with non-zero variance, were more likely to result in oscillatory behaviors, characterized by closed sensorimotor trajectories and varied motor activities.

The authors interpret their findings within the broader context of sensorimotor contingencies, a concept describing the lawful regularities that govern the interactions among sensory inputs, internal activities, and motor actions (O'Regan & Noë, 2001). This perspective holds that organisms do not passively receive sensory information, but instead actively shape their

perceptions through movements that influence their sensory experiences. By highlighting the fundamental intertwining of perception and action, the theory of sensorimotor contingencies suggests that cognitive processes are grounded in real-time interactions with the environment rather than relying on internal representations (Noë, 2004). This framework explains behavior as a dynamic construct arising from the ongoing coordination of sensory and motor activities (O'Regan & Noë, 2001).

The authors use this model to illustrate how random motor babbling can generate a sensorimotor habitat, i.e. a set of actual possible sensorimotor trajectories based on an agent's internal activity and environmental interactions. In this model, sensorimotor regularities emerge dynamically from the robot's interaction with its environment, rather than having to be pre-programmed.

The authors draw a parallel between the findings of their study and the skin-brain thesis, the view that early nervous systems evolved from contractile tissues spread across an organism's body, leading to coordinated patterns of contraction and extension (Keijzer, F., van Duijn, M., & Lyon, P., 2013). They suggest that the random motor activity in their experiment is analogous to the self-induced motility in early organisms, both contributing to the formation of sensorimotor patterns that are beneficial for biological survival. This connection between the evolution of sensorimotor patterns and the development of early nervous systems provides some potential insight into the biological origins of habits as enactivists understand them.

The study's findings reaffirm enactive theory, which views habits as self-sustaining behavioral patterns that emerge from repeated interactions between an agent and its environment. The

IDSM-based agent demonstrates habits as dynamic and emergent, given that plastic changes in the sensorimotor medium can result in the formation. The results also align with the enactive view that habits arise from sensorimotor interactions rather than being driven by internal representations.

Zarco and Egbert (2023) successfully expanded upon the IDSM framework by showing how random motor activity and parameter variations can lead to the formation of different types of sensorimotor habits. Their work reinforces the enactive approach to habits, highlighting the biological and evolutionary influences that shape sensorimotor autonomy and the emergence of habits through dynamic interactions with the environment. But the social dimension of habitual behavior remains a key feature of the enactive perspective that is yet to be discussed, and the next study presents an intriguing model in this area.

Towards modelling social habits: an organismically inspired evolutionary robotics approach.

Although previous attempts have proven successful in modeling autonomous habits, there remains a need to integrate the social aspects of habitual behavior, as these are fundamental to the enactive perspective of habits. In response to this need, Bedia et al. (2019) developed a model that incorporates the social dimensions of habits into its framework. Their study investigates how internal behavioral stability and social interaction contribute to the emergence and maintenance of social habits.

In their experiment, Bedia and colleagues artificially evolved agents to exhibit phototaxis (movement toward light) in a two-dimensional simulation. The agents were evaluated with a

single light source at a time, randomly placed at varying distances. The agents could sense each other, and they were encouraged to stay close to these randomly placed light sources. Importantly, the agents were controlled by a homeostatically plastic, 6-neuron fully connected Continuous Time Recurrent Neural Network (CTRNN), which allowed for the generation of autonomous sensorimotor behavior, as demonstrated in previous studies by Barandiaran and Di Paolo (2010).

The artificial evolution process involved two distinct fitness functions to assess the agents' performance. The primary fitness function incorporated proximity to light, time spent near the light, and the degree of homeostatic stability in the agents' neural activity (i.e., the amount of time neurons operated homeostatically without inducing plastic changes). A second fitness function omitted the homeostatic component, allowing the researchers to compare homeostatic agents with non-homeostatic agents to evaluate the role of stability in habit formation.

In one experiment, one agent was blinded by disabling its light sensors, while the other agent retained its ability to perceive the light. And the experiment showed that agents with homeostatic stability were able to collectively achieve the phototactic task even with one agent being blinded. This was possible due to the synaptic plasticity in the agent's neural controller, which, despite sensory impairment, was able to maintain a collective behavior pattern. By relying on social interaction (tracking the other agent), the blinded agent maintained homeostatic stability, and was able to retain some degree of success in performing a task.

On the other hand, in the absence of homeostatic stability, blinded agents were unable to

maintain their phototactic habits, and this disruption negatively impacted the performance of even non-blinded agents. These findings not only highlight the centrality of social perception in habit formulation, but also show how homeostatic plasticity can stabilize collective behavior and ensure that social habits persist even when individual sensory capabilities are impaired.

The findings of this paper support the idea that social habits can emerge through the same mechanisms that allow for internal behavior stability while in the presence of other agents, rather than through specific internal "social mechanisms." And, in order to explain how an agent's inability to directly perceive its task objective (in this case, the light) did not necessarily disrupt its habit (as long as its social perception remained active) the authors introduce the concept of "weak enabling sociality". Which refers to how social interaction can enable the continuation of habits when faced with sensory limitations, but not fully determine them.

In the last two subsections, I will be discussing some models that give credence to the enactive idea that habits form the basis for intelligent action. The first subsection discusses a paper that shows that habits can form the underlying mechanism behind goal oriented behavior, while the second contains an overview of some artificial life and artificial intelligence models which use habits as a basis.

Goal-Oriented Behavior with a Habit-Based Adaptive Sensorimotor Map Network (Woolford

& Egbert, 2022)

In their 2022 paper, *Goal-Oriented Behavior with a Habit-Based Adaptive Sensorimotor Map Network*, Woolford and Egbert use a novel approach to model intelligent behavior (as understood by the enactive framework) in robots. To do this, they developed a tool called Adaptive Sensorimotor Map (ASM) network, a habit-based controller that enables robots to learn and adapt their behavior through repeated interactions with their environment. They use this model to present an argument for the enactive theory that intelligence emerges from an agent's continuous sensorimotor engagement rather than from abstract reasoning or representational models.

In their experiments, Woolford and Egbert applied the ASM network to minimal cognition tasks, specifically object discrimination, without using explicit fitness functions or representational structures. The ASM network enabled a robot to distinguish between different shapes—circles and diamonds—through sensorimotor learning. The robot's behavior evolved by reinforcing successful sensorimotor patterns and inhibiting those that did not lead to task success. Over time, the robot's performance improved as it consistently adapted its behavior based on feedback from its interactions with the objects.

In this model, sensorimotor habits were formed through repeated sensorimotor trajectories that got reinforced based on their success in achieving goal-oriented tasks. The ASM network integrated these trajectories into a flexible system where behaviors evolved over time, as the agent interacted with its environment. This allowed the robot to adapt its behavior dynamically without relying on external reward systems or predefined goals.

The key innovation in their model lies in how the ASM network fosters goal-directed behavior through the reinforcement of historical trajectories. By selecting sensorimotor trajectories that match previously successful behaviors, it reinforced habits that support goal-oriented tasks. As a result, the robot's behavior became increasingly aligned with the task at hand, in this case successfully identifying and responding to different objects. Therefore ASM allowance for flexible and adaptive solutions for real world tasks, contrasts it with more traditional models that rely on pre-programmed rules or external reward mechanisms.

Woolford and Egbert's work provides a significant contribution to the field of neurorobotics by demonstrating that habit-based systems can support intelligent, goal-oriented behavior in robots. Their model shows that by grounding intelligence in sensorimotor habits, robots can autonomously develop adaptive behaviors that are both flexible and resilient in complex environments.

Agents of Habit: Refining the Artificial Life Route to Artificial Intelligence (Ramírez-Vizcaya & Froese, 2020)

In the paper Agents of Habit: Refining the Artificial Life Route to Artificial Intelligence, Susana Ramírez-Vizcaya and Tom Froese explore how habits could serve as the foundation for modeling intelligent behavior within Artificial Life (ALife) and Artificial Intelligence (AI). The authors challenge the conventional view of intelligence, which typically emphasizes higher-order cognitive functions like reasoning and problem-solving, and instead propose that habits—understood as adaptive, self-sustaining patterns of behavior—are the building blocks for mental life and intelligence.

The authors argue that ALife research offers a promising route for AI by departing from traditional cognitive science models that rely on mental representations and inferences. They propose that intelligence emerges from embodied and situated know-how rather than abstract reasoning. An agent's habits form the basis for its adaptive behavior and ability to maneuver through complex environments. Rather than internal cognitive models, an agent's sensorimotor engagements, which allow for continuous interaction between it and its environment, determine intelligent behavior.

An agent's habits are not rigid, automatic responses to stimuli, but instead are adaptive and flexible, dynamically adjusting to changes in the environment while maintaining the agent's behavioral stability. This flexibility allows for both the persistence of habits and their transformation over time, enabling an agent to develop an increasingly complex network of habits. According to Vizcaya and Froese, this capacity for adapting and refining habits is essential for scaling up from basic sensorimotor behaviors to more complex forms of intelligent behavior, such as decision-making and problem-solving.

Finally, the authors also highlight the limitations of current enactive models of habits. These models are either too disembodied, allowing for a rich but detached landscape of possible behaviors, or too minimal, focusing on embodied agents but restricting the complexity of their habits. To address this, they suggest bridging these approaches by developing models that incorporate both embodied, situated agents and a rich network of interrelated habits. They argue that the key to advancing AI lies in creating agents capable of developing a complex, self-organizing web of habits, which could ultimately give rise to intelligent behavior.

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